Phase 1: Data Exploration, Preprocessing and Retrieval Preparation

MS COCO 2017 Dataset

Groupe 08

1. Introduction

Image captioning is a task in artificial intelligence that involves generating natural-language descriptions of images. It combines two important fields: computer vision, which enables a computer to interpret visual information, and natural language processing, which allows it to express that information through text.

This project focuses on preparing the data and technical resources needed to develop an image captioning system using the MS COCO 2017 dataset.

The first phase is dedicated to understanding and preparing the dataset before developing the complete system. It covers dataset exploration, image and annotation verification, statistical analysis, data cleaning, image preprocessing, and the preparation of image and text representations for retrieval.

The work was divided among four team members, with each member responsible for a specific part of the preparation process. Their contributions were then combined to create a shared dataset and establish a common foundation for the next phase of the project.

2. Project Objectives

The main objective of this phase is to prepare a clean, organized, and well-documented dataset that can be used in the development of the image captioning and retrieval components.

The specific objectives are to:

- Explore the structure and contents of the MS COCO 2017 dataset.
- Select the images and annotations required for the project.
- Check image integrity and annotation consistency.
- Identify missing, corrupted, and duplicate images.
- Analyze object categories, caption characteristics, and scene complexity.
- Generate statistical graphs to better understand the dataset.
- Organize the images into knowledge-base, validation, and test subsets.
- Configure and test image preprocessing using CLIP.
- Extract and organize image, object, and caption metadata.
- Generate image and text embeddings.
- Build FAISS indexes to support similarity-based retrieval.
- Perform quality-control checks and preliminary retrieval experiments.

3. Dataset Description

3.1. MS COCO 2017

The project uses MS COCO 2017 (Microsoft Common Objects in Context), a dataset widely used in computer vision research.

It contains images of everyday objects and real-world scenes. Each image can be associated with object annotations, bounding boxes, and natural-language captions.

The dataset includes 80 object categories, such as person, car, bicycle, dog, chair, and many others.

Two main types of annotations are used in this project:

- **Object annotations:** Describe the objects present in an image, including their categories and bounding-box coordinates.
- **Caption annotations:** Provide human-written descriptions of the image content.

These annotations provide both visual and textual information, making the dataset suitable for image captioning and retrieval experiments.

3.2. Dataset Structure

Two subsets of MS COCO 2017 were used during this phase:

- **train2017:** Used to select the knowledge-base and validation images.
- **val2017:** Used to prepare the independent test set.

The main annotation files are listed below.

File name	Description
instances_train2017.json	Contains object annotations and bounding boxes for training images.
captions_train2017.json	Contains captions associated with training images.
captions_val2017.json	Contains captions associated with validation images.

The selected images were organized into three subsets:

Subset	Number of images	Source
Knowledge Base	10,000	train2017
Validation	100	train2017
Test	100	val2017
Total	**10,200**	

The test subset was selected separately from val2017 to keep it independent of the knowledge-base and validation images.

3.3. Initial Dataset Statistics

Before selecting the images required for the project, the original train2017 annotation files were examined.

The initial dataset contained:

Dataset information	Quantity
Images	118,287
Object annotations	860,001
Object categories	80
Caption annotations	591,753

These figures describe the original annotation files, not the smaller subset selected for the project.

4. Development Environment and Technologies

The project was developed using Python in Google Colab. Google Drive was used as shared storage, allowing all team members to access the same dataset, notebooks, and generated files.

The following tools and libraries were used:

Technology	Purpose
Python	Main programming language used throughout the project.
Google Colab	Environment for developing and executing Python notebooks.
Google Drive	Shared storage and collaboration between team members.
Pandas	Organizing, processing, and analyzing structured data.
NumPy	Numerical calculations and embedding handling.
Matplotlib	Generating statistical graphs and visualizations.
Pillow (PIL)	Reading, checking, and preprocessing images.
JSON	Loading and saving annotation and metadata files.
hashlib	Calculating SHA-256 hashes for duplicate detection.
Hugging Face Transformers	Loading the CLIP model and its processor.
PyTorch	Working with tensors and computing embeddings.
FAISS	Creating and searching vector indexes.

These tools were used together to explore and prepare the dataset, analyze its contents, generate numerical representations, and perform initial retrieval experiments.

The project files were organized in a shared Google Drive folder named COCO_Project. The folder contains five main directories:

- **annotations/** – stores the COCO annotation files and prepared annotation files.
- **images/** – contains the selected image datasets.
- **metadata/** – stores image identifiers, dataset manifests, and preprocessing configurations.
- **statistics/** – contains statistical results and generated visualizations.
- **test/** – contains the separate test dataset.

This organization made it easier for team members to access the project resources and keep the different types of files separate.

5. Part 1 — Dataset Exploration and Integrity Verification

5.1. Exploration of the Original Dataset

The first stage involved exploring the original MS COCO 2017 dataset and examining its annotation files.

Two main files were used:

- **instances_train2017.json:** contains image information, object annotations, bounding boxes, and object categories.
- **captions_train2017.json:** contains image information and their corresponding textual descriptions.

The initial examination produced the following results:

Dataset element	Quantity
Images in the training annotations	118,287
Object annotations	860,001
Object categories	80
Caption annotations	591,753

This initial exploration provided an overview of the available data and helped determine which information was needed for the project.

5.2. Selection of the Project Dataset

A subset of the original COCO training dataset was selected using Python. A fixed random seed of 42 was used to make the selection reproducible.

A total of 10,100 images were selected from train2017 and divided into two subsets:

- **Knowledge Base:** 10,000 images used as the main collection for image and text retrieval.
- **Validation Set:** 100 images reserved for validation.

The selected image identifiers and their corresponding filenames were saved in metadata/selected_image_ids.csv.

A separate test set containing 100 images was also selected from the COCO 2017 validation dataset (val2017). This test set was kept independent of the training and validation subsets.

5.3. Image Integrity Verification

After selecting the images, several checks were performed to make sure that the files were available and suitable for further processing.

The verification process included:

- Checking whether all selected image files existed.
- Opening and decoding the images to detect corrupted files.
- Calculating SHA-256 hashes to identify exact duplicate images.
- Checking for overlapping image identifiers between the knowledge base and validation subsets.

The audit confirmed that all 10,100 selected training images were available and valid. No corrupted images, missing files, or exact duplicate groups were detected.

The results are summarized below:

Integrity check	Result
Total images checked	10,100
Missing images	0
Corrupted images	0
Exact duplicate groups	0
Duplicate image IDs	0
Overlap between knowledge base and validation	0

These results indicate that the selected images passed the main integrity checks.

5.4. Annotation Consistency Verification

The selected image identifiers were compared with the identifiers available in both annotation files. The purpose was to make sure that each selected image could be matched with its corresponding captions and object annotations.

The results were:

Annotation check	Result
Images missing from instance annotations	0
Images missing from caption annotations	0
Images without object annotations	104
Images without captions	0

The 104 images without object annotations were retained because each of them had five associated captions.

Since the main objective of the project is image captioning, the absence of object annotations does not automatically make these images unusable. Their captions still provide useful textual information about the image content.

5.5. Results and Generated Files

The dataset exploration and integrity verification confirmed that the selected training subset was suitable for the following analysis and preprocessing stages.

The main files generated during this stage were:

- **metadata/selected_image_ids.csv** – contains the selected image identifiers and their assigned subsets.
- **metadata/dataset_audit_report.json** – summarizes the dataset integrity and annotation consistency results.
- **metadata/corrupted_images.csv** – records corrupted images detected during verification.
- **metadata/duplicate_images.csv** – records exact duplicate images detected during verification.

Both the corrupted-image and duplicate-image reports were empty, confirming that no such cases were detected in the selected 10,100 images.

Overall, this stage established a verified dataset and provided the information needed for statistical analysis, preprocessing, and retrieval preparation.

6. Part 2 — Statistical Analysis and Data Visualization

6.1. Objective of the Statistical Analysis

After verifying the integrity of the selected dataset, statistical analysis was performed to better understand its characteristics and distribution.

The analysis focused on image dimensions, image orientation, caption length, object categories, and the number of objects appearing in each image.

Pandas and NumPy were used to process the data and calculate statistical measures, while Matplotlib was used to generate visualizations.

6.2. Image Dimensions and Orientation

The dimensions of the 10,100 selected images were examined to understand their size distribution and orientation.

The results are presented below:

Measurement	Width (pixels)	Height (pixels)
Mean	576.77	483.27
Standard deviation	92.53	97.64
Minimum	120	128
Median	640	480
Maximum	640	640

The images were also classified according to their orientation:

Orientation	Number of images
Landscape	7,328
Portrait	2,398
Square	374
Total	**10,100**

The results show that landscape images represent the largest proportion of the selected dataset (Figure 1).

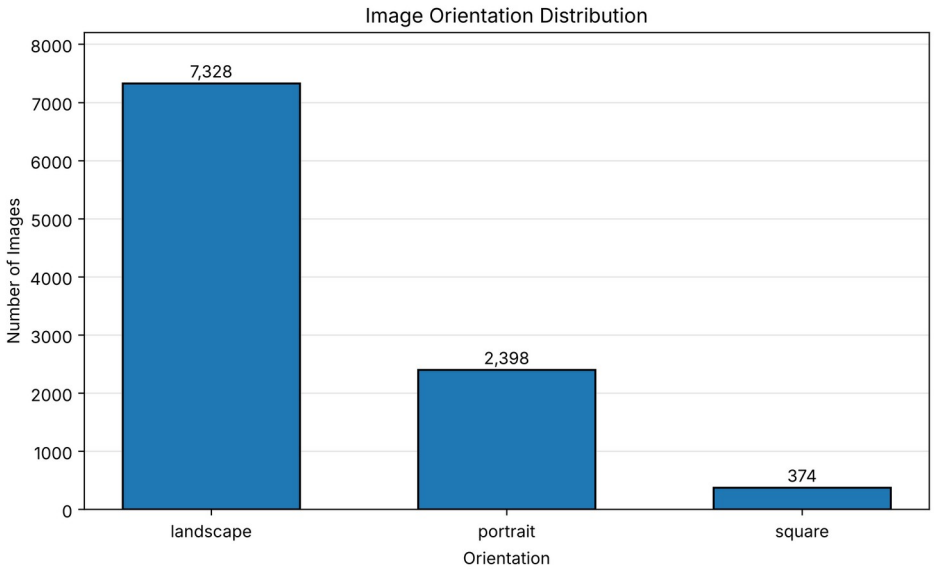


Figure 1: Distribution of image orientations in the 10,100 selected images.

6.3. Caption Length Analysis

The captions associated with the selected images were analyzed to examine their length and distribution.

A total of 50,521 captions were included in the analysis.

The main statistical results are:

Statistical measure	Result
Total captions	50,521
Mean caption length	10.46 words
Median caption length	10 words
Minimum length	6 words
Maximum length	47 words

The results show that captions are generally short, with an average length of approximately ten words (Figure 2). This gives an idea of the type of textual descriptions that the image captioning system will process.

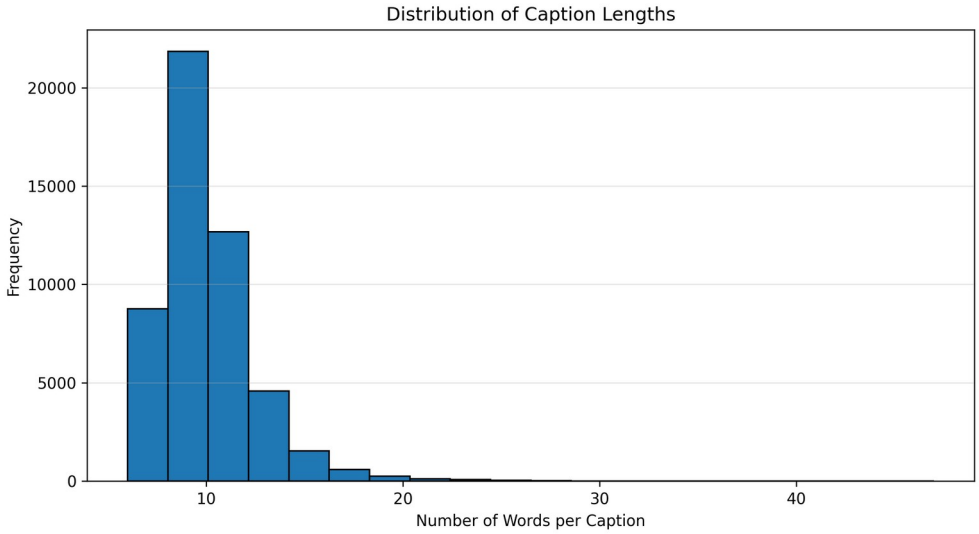


Figure 2: Distribution of caption lengths (number of words per caption).

6.4. Object Category Distribution

The object annotations were analyzed to identify the categories that appear most frequently in the selected dataset.

A total of 72,752 object annotations were identified across the 10,100 selected images, covering all 80 COCO object categories.

The ten most frequent categories were:

Object category	Number of annotations
Person	22,319
Car	3,664
Chair	3,207
Bottle	2,064
Book	2,009
Cup	1,672
Dining table	1,287
Bowl	1,235
Umbrella	1,097
Handbag	1,031

The person category is clearly the most frequent in the selected dataset (Figure 3). Everyday objects and transportation-related objects are also well represented.

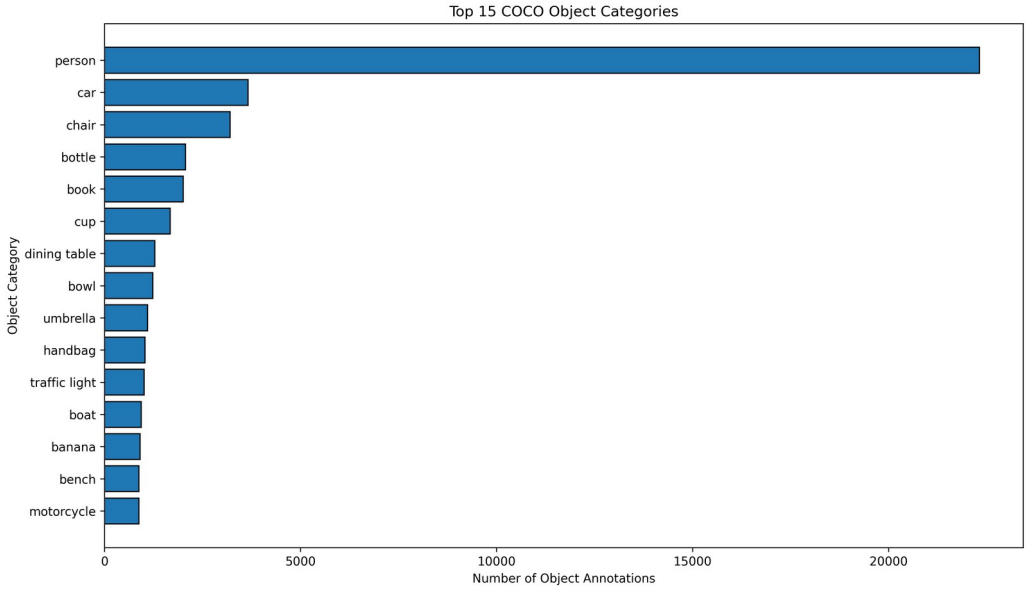


Figure 3: The 15 most frequent object categories in the selected dataset.

6.5. Scene Complexity Analysis

Scene complexity was examined by calculating the number of object annotations associated with each image.

The analysis produced the following descriptive statistics:

Statistical measure	Number of objects
Mean	7.20
Standard deviation	7.21
Minimum	0
First quartile (Q1)	2
Median	4
Third quartile (Q3)	10
Maximum	56

The images were also grouped into four categories according to their number of annotated objects:

Number of objects	Number of images
0 objects	104
1–5 objects	5,734
6–10 objects	1,897
More than 10 objects	2,365
Total	**10,100**

Most images contain between one and five annotated objects. However, a considerable number contain more than ten objects, showing that the dataset includes scenes with different levels of visual complexity (Figure 4).

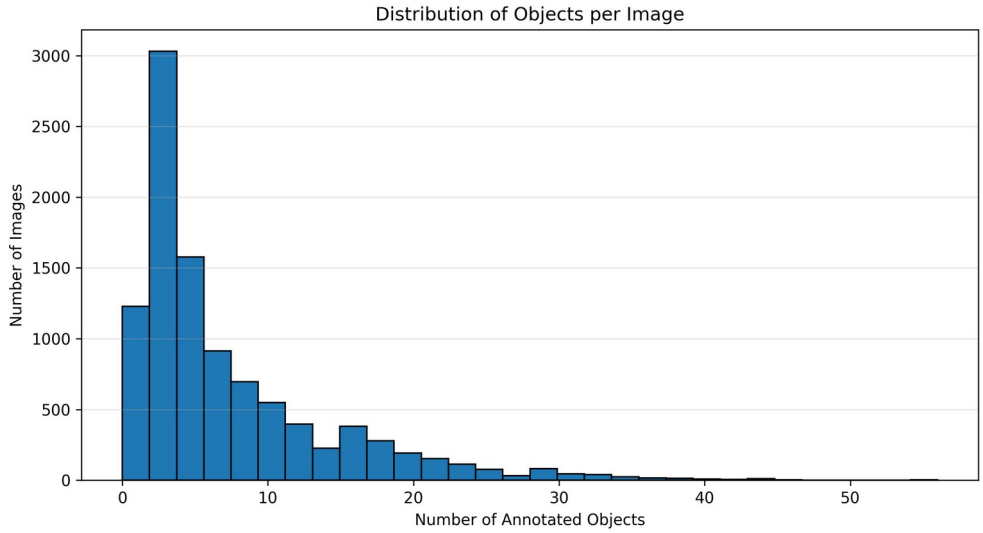


Figure 4: Distribution of the number of annotated objects per image.

6.6. Data Visualization

Matplotlib was used to create visual representations of the statistical results. These graphs make it easier to identify patterns, compare distributions, and understand the characteristics of the dataset.

The following visualizations were generated:

- **Image Orientation** (Figure 1): compares the number of landscape, portrait, and square images.
- **Caption Length Distribution** (Figure 2): illustrates the distribution of caption lengths.
- **Object Category Distribution** (Figure 3): displays the frequency of the different object categories.
- **Scene Complexity** (Figure 4): shows how images are distributed according to their number of annotated objects.
- **Caption Attribute Distributions** (Figures 5 to 7): show the color, part-of-speech, and shape and size words found in the captions.
- **Split Distribution** (Figure 8, Section 7.2): shows the distribution of images across the selected subsets.

The generated figures are stored in the statistics/ directory.

6.7. Caption Linguistic Analysis

In addition to caption length, the content of the captions was examined to understand which kinds of words describe the images. Three aspects were analyzed: color words, part-of-speech categories, and words describing shape and size. The values below are approximate and are read from the generated charts.

Color words: white is by far the most frequent color mentioned in the captions, with about 3,100 occurrences, followed by black (about 1,700) and red (about 1,450). Blue and green appear roughly 1,000 to 1,200 times each, while purple is the least frequent color (about 140 occurrences) (Figure 5).

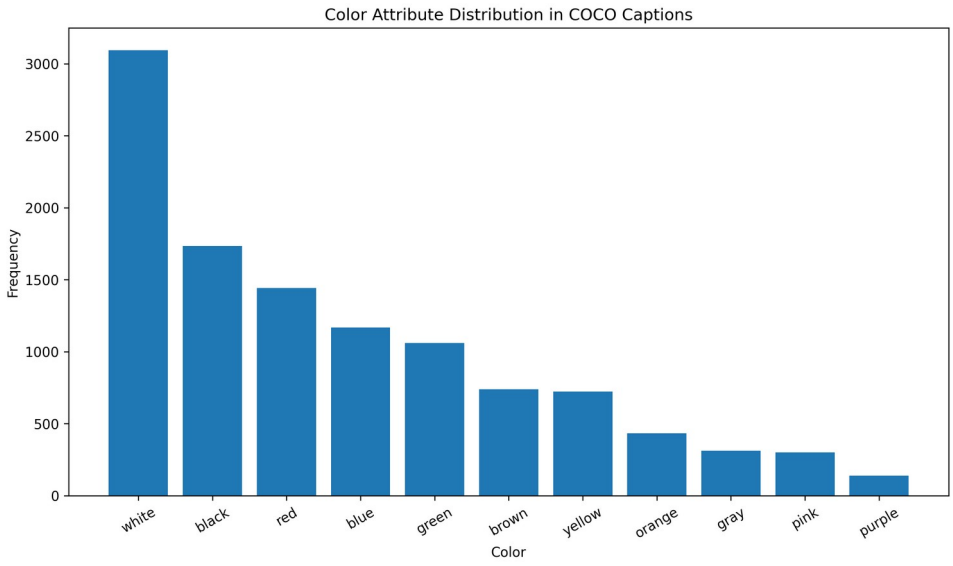


Figure 5: Frequency of color words in the captions.

Part-of-speech distribution: nouns make up about one third of the words in the captions (around 33%), followed by verbs (about 11%) and adjectives (about 8%). Adverbs are very rare (under 1%), and the remaining words (about 48%) fall into the other category (Figure 6). The high share of nouns reflects the object-centered nature of the descriptions.

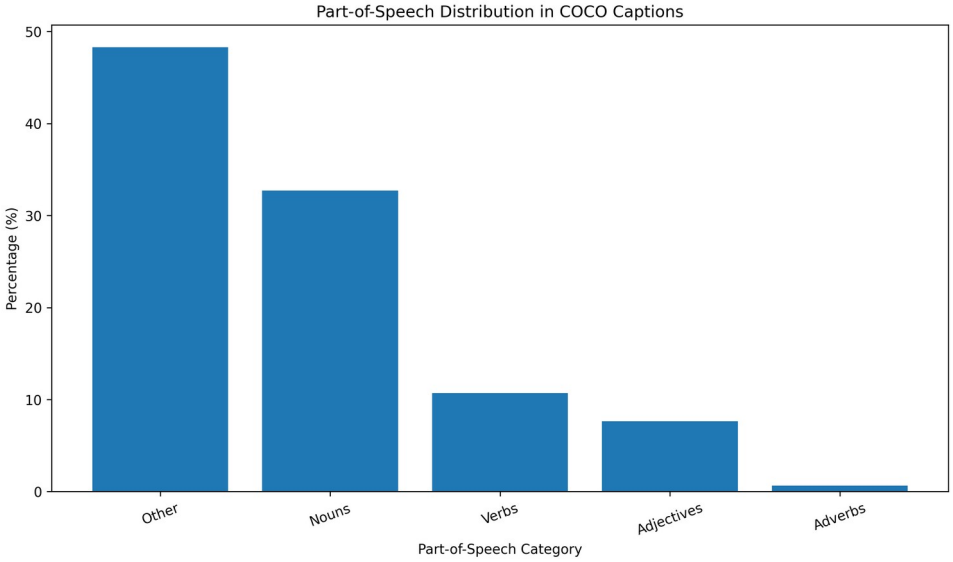


Figure 6: Part-of-speech distribution of the words in the captions.

Shape and size words: size words dominate this group, with "large" (about 2,050 occurrences) and "small" (about 1,670) far ahead of the others. "Long" comes next (about 280), while the words describing geometric shapes, such as square, round, or circular, each appear fewer than 100 times (Figure 7).

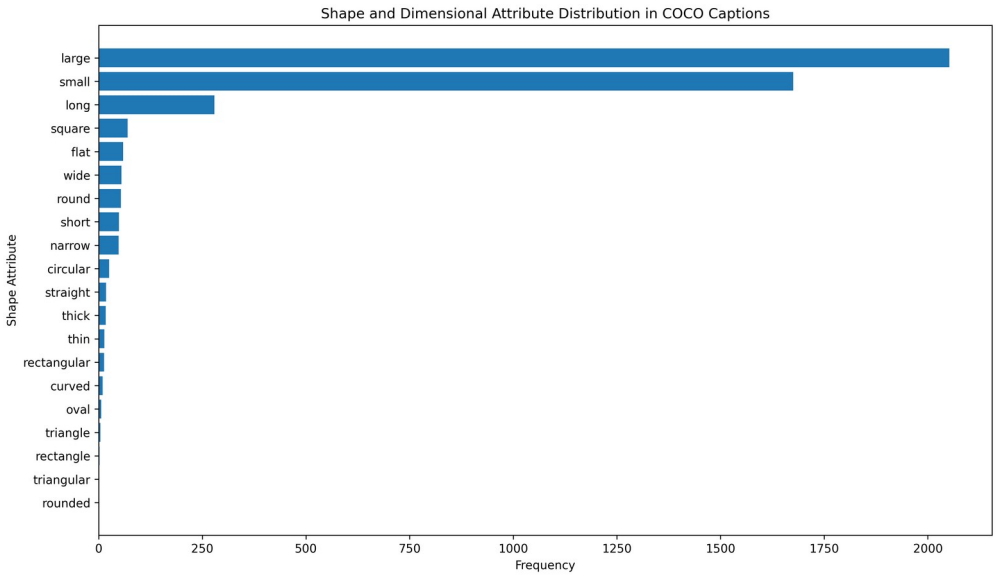


Figure 7: Frequency of shape and size words in the captions.

6.8. Generated Statistical Files

Several files were saved to preserve the analysis results and make them available to the other project members.

The main files include:

- metadata/caption_analysis.csv
- metadata/object_category_counts.csv
- metadata/scene_complexity.csv
- metadata/clean_dataset_manifest.csv
- metadata/test100_manifest.csv

These files contain structured statistical information and dataset records that can be reused in the following stages.

6.9. Summary

The statistical analysis provided a clearer picture of the selected dataset. It contains 10,100 images, 50,521 captions, and 72,752 object annotations.

The results also showed that most images are landscape-oriented, captions generally contain around ten words, and the number of objects varies considerably from one image to another.

These findings helped the team better understand the visual and textual characteristics of the dataset before moving on to image preprocessing, feature extraction, and retrieval.

7. Part 3 — Data Cleaning, Dataset Splitting and Image Preprocessing

7.1. Objective

The objective of this stage was to prepare a clean and organized dataset for the next phases of the project.

This involved organizing the selected images into separate subsets, checking their quality, preparing the test dataset, and applying the image preprocessing operations required by the CLIP model.

7.2. Dataset Organization and Splitting

The selected images were organized into three independent subsets:

Subset	Number of images	Purpose
Knowledge Base	10,000	Main image collection used for retrieval
Validation	100	Used for validation and testing the processing pipeline
Test	100	Reserved for independent evaluation
Total	**10,200**	

The knowledge-base and validation images were selected from COCO 2017 train2017, while the independent test images were selected from val2017.

A random seed of 42 was used during selection to ensure reproducibility. Image identifiers were checked to confirm that there was no overlap between the three subsets.

The validation and test datasets were kept separate from the knowledge base to avoid mixing evaluation data with the main retrieval collection.

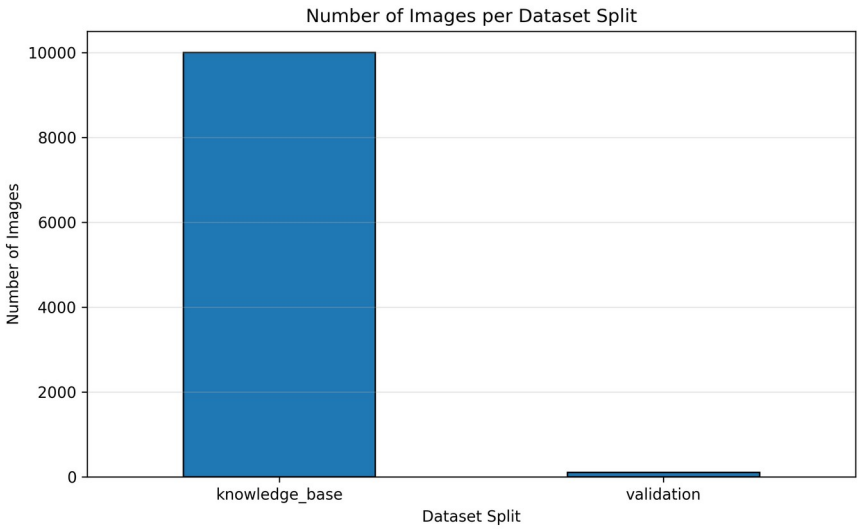


Figure 8: Number of images per dataset split. The test set is not shown because its 100 images come from val2017 and are stored separately.

7.3. Test Dataset Preparation

A separate test dataset was created using images from the COCO 2017 validation split.

The annotation file captions_val2017.json was used to retrieve the reference captions associated with the selected test images. A total of 100 unique images were selected, each with five reference captions.

The final test dataset contained:

Test dataset information	Result
Images	100
Reference captions	500
Images missing captions	0
Invalid images	0
Unique image identifiers	100

All downloaded images were checked to confirm that they were valid JPEG files.

During downloading, the original COCO image server produced an SSL hostname verification error. To complete the download, an alternative Hugging Face mirror was used. The downloaded files were then verified to make sure that the download had succeeded correctly.

The test images were stored in test/val2017/, while their annotations and identifiers were saved in the corresponding project metadata and annotation files.

7.4. Image Cleaning and Dataset Verification

Before preprocessing, the selected images were checked to make sure that they were available and could be opened successfully.

The cleaning results showed that:

- No missing images were found.
- No corrupted images were detected.
- No exact duplicate image groups were identified.
- All selected images had associated captions.

The 104 images without object annotations were retained because they still contained valid captions. This decision was based on the project's focus on image captioning rather than object detection alone.

The original image files were preserved, and preprocessing was performed without modifying the source images.

7.5. Image Preprocessing Using CLIP

To prepare the images for feature extraction, the project used the pretrained CLIP model openai/clip-vit-base-patch32.

The CLIPImageProcessor provided by the Hugging Face Transformers library was used to apply the preprocessing operations expected by the model.

The preprocessing pipeline consisted of five main steps:

- **RGB conversion:** Images were converted to RGB format to ensure consistent color channels.
- **Resizing:** The shortest image edge was resized to 224 pixels.
- **Center cropping:** A central region of 224 × 224 pixels was extracted.
- **Rescaling:** Pixel values were rescaled to the expected numerical range.
- **Normalization:** The image channels were normalized using the CLIP model's predefined mean and standard deviation values.

The normalization parameters were:

Channel	Mean	Standard deviation
Red	0.48145466	0.26862954
Green	0.45782750	0.26130258
Blue	0.40821073	0.27577711

These operations ensure that the images follow the input requirements of the pretrained CLIP model.

7.6. Preprocessing Verification

After configuring the preprocessing pipeline, a sample image was processed to verify that the operations were working correctly.

The test produced a tensor with the following properties:

Property	Result
Input image size	640 × 480 pixels
Processed image size	224 × 224 pixels
Tensor shape	(1, 3, 224, 224)
Data type	torch.float32
Processing status	Successful

The result confirmed that the preprocessing pipeline produced an input tensor compatible with the CLIP image encoder.

The preprocessing configuration was saved in metadata/clip_preprocessing_config.json. This allows the same settings to be reused in later stages.

7.7. Generated Files and Outputs

The main outputs of this stage included:

- **metadata/clean_dataset_manifest.csv** – contains information about the cleaned dataset.
- **metadata/test100_manifest.csv** – contains the identifiers and information for the 100 test images.
- **metadata/clip_preprocessing_config.json** – stores the CLIP preprocessing configuration.
- **annotations/captions_test100.json** – contains the reference captions for the selected test images.
- **test/val2017/** – contains the downloaded test images.

7.8. Summary

This stage produced three organized and independent subsets containing 10,000 knowledge-base images, 100 validation images, and 100 test images.

The test dataset was downloaded and verified, and the CLIP preprocessing pipeline was successfully tested. The original images were preserved, while the preprocessing configuration was saved for consistent use in the next stage.

The resulting dataset was therefore prepared for metadata extraction, embedding generation, and vector-based retrieval.

8. Part 4 — Metadata Extraction, Embeddings and Vector Database

8.1. Objective

The objective of this stage was to prepare the dataset for image and text retrieval.

This involved extracting structured metadata, generating image and text embeddings using a pretrained CLIP model, creating vector indexes with FAISS, and performing quality-control checks.

These operations provide the technical foundation for retrieving images or captions based on visual and textual similarity.

8.2. Metadata Extraction

The first step consisted of extracting structured information associated with the selected images and captions.

A total of 10,100 metadata records were generated. These records organize information about the dataset and make it easier to identify and retrieve the corresponding images and annotations.

The metadata also supports the association between image files, captions, dataset subsets, and their generated embeddings.

8.3. Image and Text Embedding Generation

To represent images and captions numerically, the project used the pretrained model openai/clip-vit-base-patch32.

CLIP (Contrastive Language–Image Pre-training) is a multimodal model that produces vector representations for images and text. These representations can be compared to measure semantic similarity.

The model was used to generate the following embeddings:

Dataset subset	Image embeddings	Text embeddings
Knowledge Base	10,000	50,021
Validation	100	500
Test	100	Not reported

Each generated embedding had a dimension of 512, meaning that every image or text representation was stored as a vector containing 512 numerical values.

The image and text embeddings allow the retrieval system to compare visual and textual information within a shared representation space.

8.4. Vector Indexing with FAISS

After generating the embeddings, the project used FAISS (Facebook AI Similarity Search) to organize the knowledge-base vectors and support efficient similarity searches.

Two separate vector indexes were created:

Index	Number of vectors
Image FAISS index	10,000
Text FAISS index	50,021

The image index stores the knowledge-base image embeddings, while the text index stores the knowledge-base text embeddings.

These indexes allow the system to search for the most similar vectors efficiently.

8.5. Retrieval Performance

Retrieval experiments were conducted to evaluate the performance of the vector search system.

The reported average retrieval latency for a top-10 search was 0.931 milliseconds per query.

This result indicates that the FAISS indexes can return a ranked list of similar vectors with low search latency under the tested conditions.

The top-10 retrieval setting means that the system retrieves the ten nearest results for a given query.

8.6. Quality-Control Checks and Test Data Isolation

Quality-control checks were performed to verify the generated metadata, embeddings, and vector indexes. According to the reported results, all quality-control checks passed.

The test-image embeddings were generated separately and were not added to the knowledge-base FAISS indexes. This separation is important because it prevents test images from becoming part of the retrieval collection used during evaluation.

Keeping the test data separate helps reduce the risk of data leakage and supports a more reliable evaluation process.

8.7. Results of Part 4

The main results of this stage can be summarized as follows:

Result	Quantity or status
Structured metadata records	10,100
Knowledge-base image embeddings	10,000
Knowledge-base text embeddings	50,021
Validation image embeddings	100
Validation text embeddings	500
Test-image embeddings	100
Embedding dimension	512
Image FAISS index	10,000 vectors
Text FAISS index	50,021 vectors
Average top-10 retrieval latency	0.931 ms/query
Quality-control checks	Passed

8.8. Summary

This stage transformed the prepared dataset into structured metadata and numerical image and text representations. The generated embeddings were organized into FAISS indexes to support similarity-based retrieval.

The reported retrieval latency and successful quality-control checks demonstrate that the retrieval preparation stage was completed successfully. Furthermore, keeping test-image embeddings outside the knowledge-base indexes helps preserve the separation between the retrieval collection and the independent test set.

9. Overall Results and Discussion

9.1. Summary of Project Results

The four project parts were combined to produce a prepared dataset and a retrieval-ready collection based on MS COCO 2017.

The main results obtained during Phase 1 are summarized below:

Component	Main result
Original dataset exploration	118,287 training images, 860,001 object annotations, 591,753 captions, and 80 categories
Selected training data	10,000 knowledge-base images and 100 validation images
Independent test data	100 images from COCO 2017 val2017
Image integrity	0 missing images, 0 corrupted images, and 0 exact duplicate groups
Statistical analysis	Image dimensions, orientation, caption length, caption word categories, object categories, and scene complexity analyzed
Image preprocessing	CLIP preprocessing pipeline tested successfully
Metadata extraction	10,100 metadata records generated
Image embeddings	10,000 knowledge-base image embeddings
Text embeddings	50,021 knowledge-base text embeddings
Vector indexes	Two FAISS indexes created
Embedding dimension	512
Retrieval latency	0.931 ms per top-10 query
Quality control	All reported checks passed

9.2. Discussion of Dataset Quality

The integrity verification results indicate that the selected images were successfully downloaded and could be opened and processed. No missing files, corrupted images, or exact duplicate groups were detected.

The annotation analysis also confirmed that all selected images had corresponding captions. However, 104 images did not have object annotations. These images were retained because

they still contained five captions each and remained relevant to the image captioning objective.

The statistical analysis revealed that the dataset contains a variety of image orientations, caption lengths, object categories, and scene complexity levels. This variety provides different visual and textual examples for the development of the image captioning system.

9.3. Discussion of Preprocessing and Retrieval Preparation

The preprocessing stage successfully prepared the images according to the input requirements of the CLIP model. The test output confirmed that an image could be converted into a tensor with the expected shape of (1, 3, 224, 224).

The embedding generation stage produced numerical representations for the knowledge-base images and their associated text descriptions. These vectors were then organized into separate FAISS indexes.

The reported top-10 retrieval latency of 0.931 milliseconds per query indicates fast vector search under the tested conditions. However, this measurement should be interpreted in the context of the hardware, index configuration, and experimental setup used during testing.

9.4. Overall Achievements

At the end of Phase 1, the team successfully completed the main data preparation and retrieval tasks.

The main achievements were:

- Exploring the original MS COCO 2017 dataset and its annotation structure.
- Selecting and verifying the required image subsets.
- Analyzing the statistical characteristics of the dataset.
- Preparing the images using a consistent CLIP preprocessing pipeline.
- Creating structured metadata for the selected images.
- Generating image and text embeddings.
- Building separate FAISS indexes for image and text retrieval.
- Performing quality-control checks and keeping test images separate from the knowledge-base indexes.

Together, these results establish the data preparation and retrieval foundation required for the subsequent development of the image captioning system.